



Complete Paper Solution

EGD – Summer 2023

Engineering Graphics and Design

(GTU)



Q.1 (a) Define the functions with symbols: 1) Continuous thin, 2) Chain thin and thick at ends and 3) Chain thin. **(03 Marks)**

1) Continuous thin: _____

This type of line is used as extension lines, dimensions lines, leader lines, construction lines, projection lines, hatching lines etc.

2) Chain thin and thick at ends: _____

This type of line is used as cutting plane line.

3) Chain thin: _____

This type of line is used as center lines, locus lines, pitch lines and pitch circle.



Q. 1 (b) Construct a scale to measure km, $\frac{1}{8}$ km and $\frac{1}{40}$ of a km, in which 1 km is represented by 4 cm. Mark on this scale a distance of 2.775 km. (04 Marks)

Given data

Actual length = 1 km

Drawing length = 4 cm

To show = km, $\frac{1}{8}$ km & $\frac{1}{40}$ km

To measure = 2.775 km

Maximum length = Assume 3 km

$$R.F. = \frac{\text{Drawing length}}{\text{Actual length}}$$

$$= \frac{4 \text{ cm}}{1 \text{ km}}$$

$$= \frac{4 \text{ cm}}{1 \times 1000 \times 100 \text{ cm}}$$

$$R.F. = \frac{1}{25000}$$

$$\begin{aligned} \text{Actual length of scale} &= R.F. \times \text{Max. measurement} \\ &= \frac{1}{25000} \times 3 \text{ km} \end{aligned}$$

$$\therefore A.L.S. = \frac{1}{25000} \times 3 \times 1000 \times 100 \text{ cm}$$

$$A.L.S. = 12 \text{ cm}$$

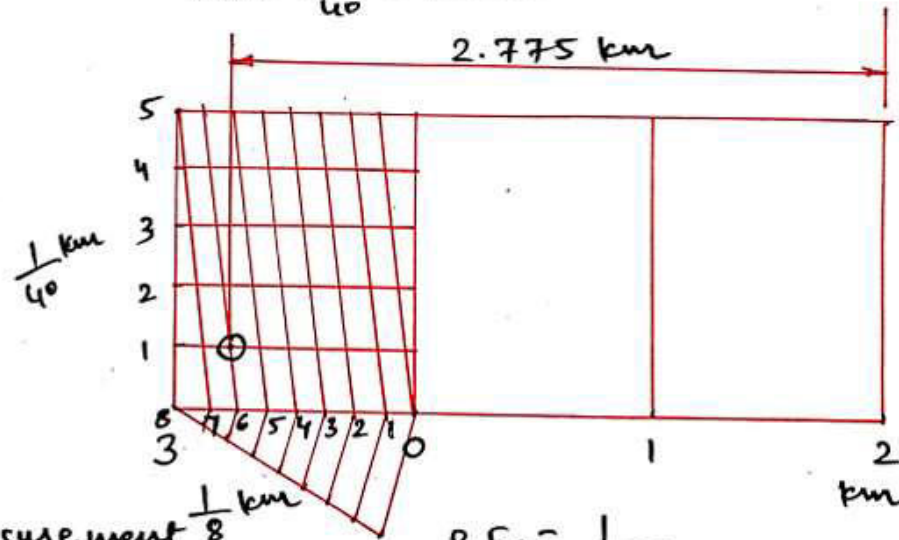
$$\frac{1}{8} = 0.125$$

$$= 0.125 \times 6 = 0.75$$

$$\text{Now } 2 + 0.75 = 2.75$$

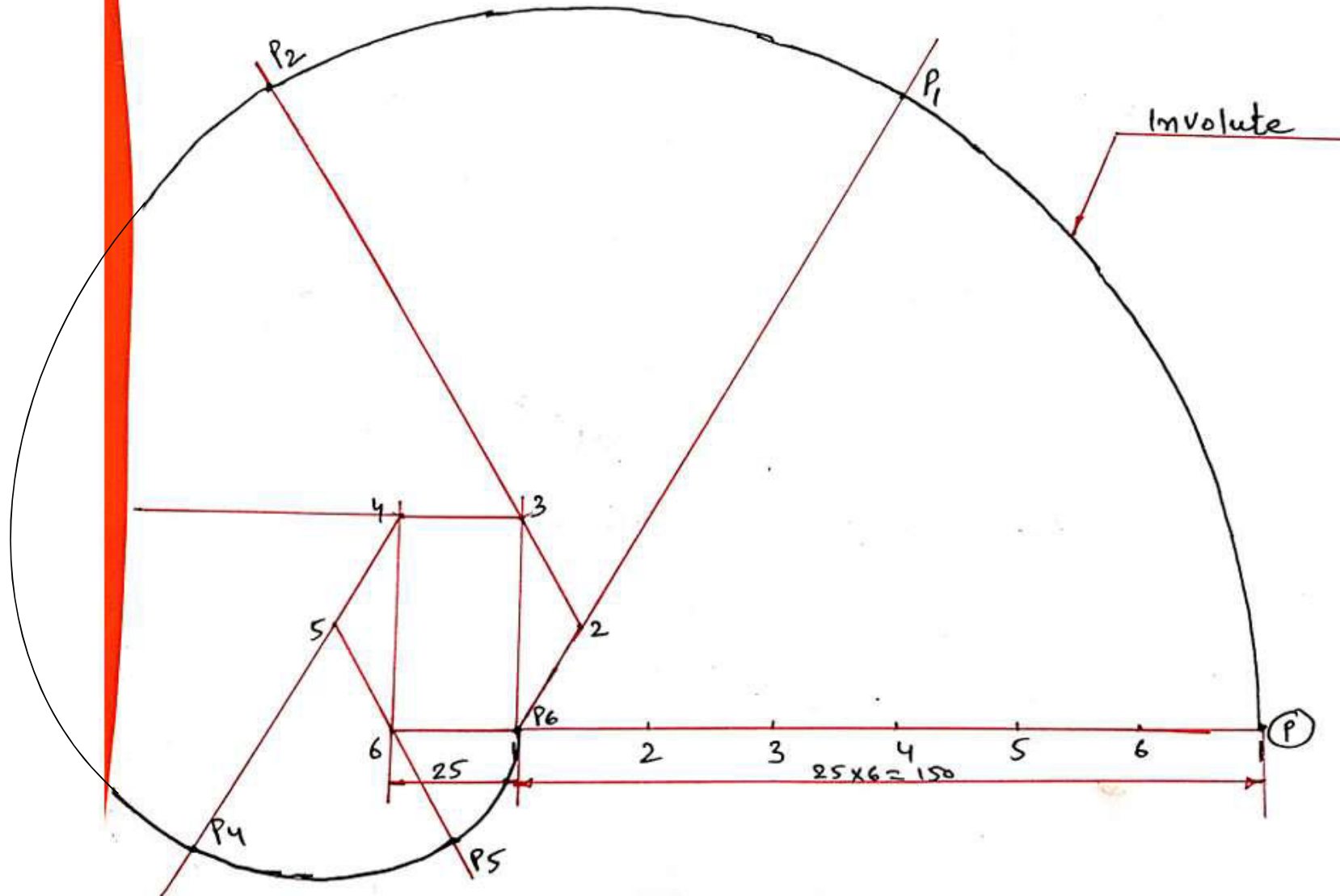
$$\text{Now } 2.775 - 2.75 = 0.025$$

$$\text{Now } \frac{1}{40} = 0.025$$



$$R.F. = \frac{1}{25000}$$

Q. 1 (c) Construct the Involute of hexagon of 25 mm sides. (07 Marks)

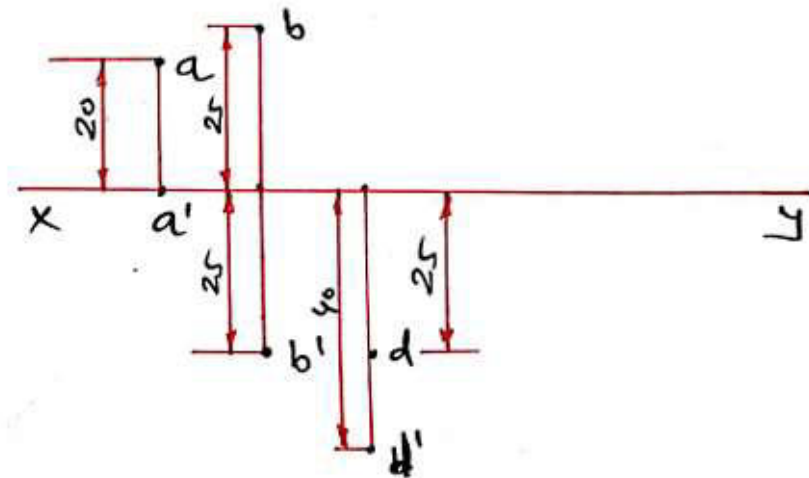


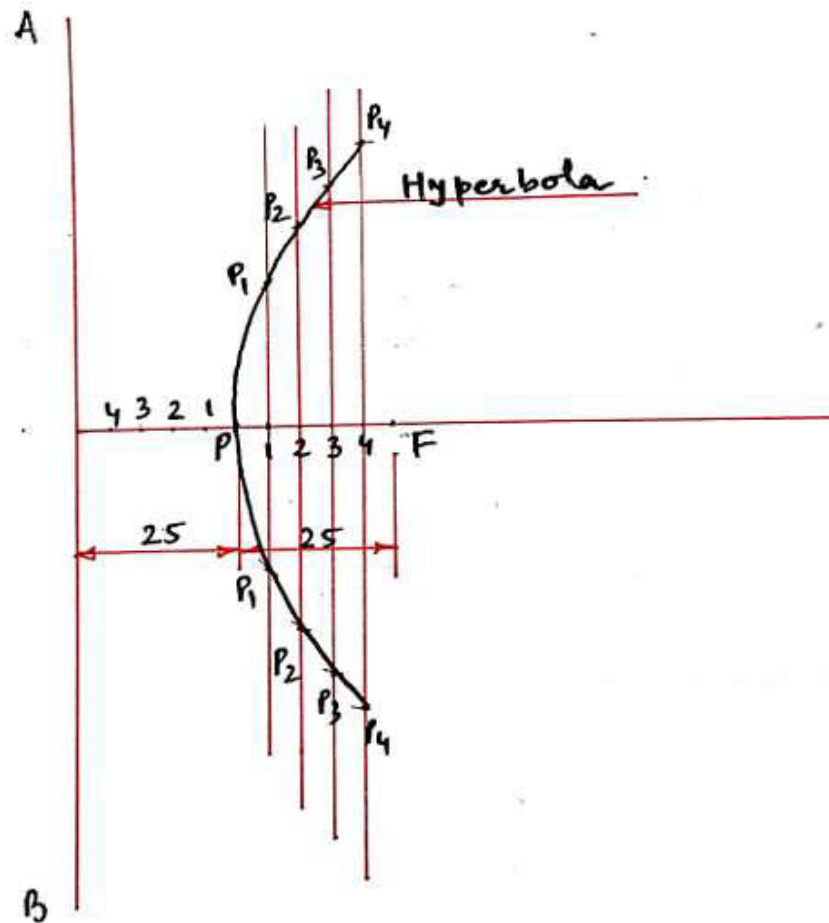


Q.2 (a) Draw the projections of the following points on the same ground lines, keeping the projectors 15 mm apart: **(03 Marks)**

- (1) A in the H.P. and 20 mm behind V.P.
- (2) B 25 mm below the H.P. and 25 mm behind V.P.
- (3) D 40 mm below H.P. and 25 mm in front of V.P.

Point A above H.P.	
Point A below H.P.	
Point A on/in the H.P.	
Point A behind V.P.	
Point A in front of V.P.	
Point A on/in the V.P.	







Q.2 (c) A wheel rolls over the horizontal straight road and covers 220 mm distance in one rotation. Draw the path traced by the point P which is initially at the point of contact between the wheel and the road. (07 Marks)

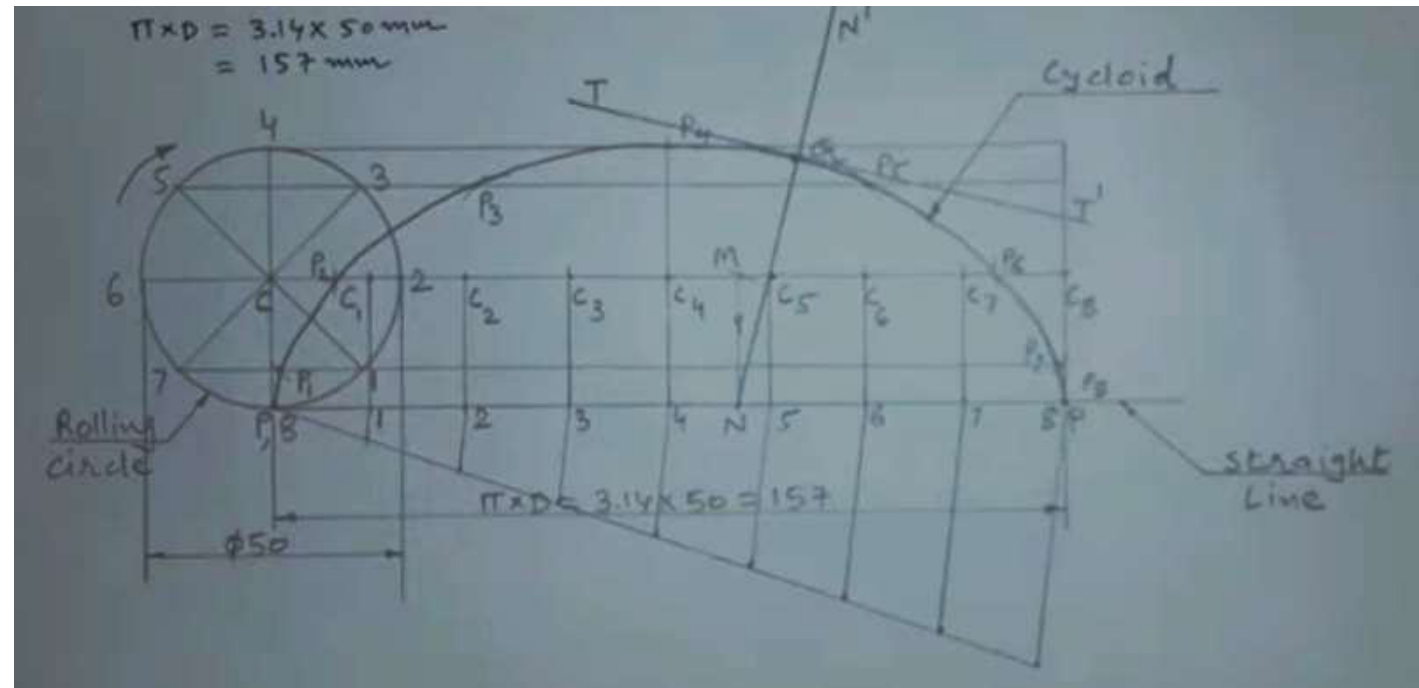
Circumference of Circle = $\pi \times D$

$$\therefore 220 = 3.14 \times D$$

$$\therefore \frac{220}{3.14} = D$$

$$\therefore D = 70\text{mm}$$

Video Link: <https://youtu.be/ukuCJLdZckw>



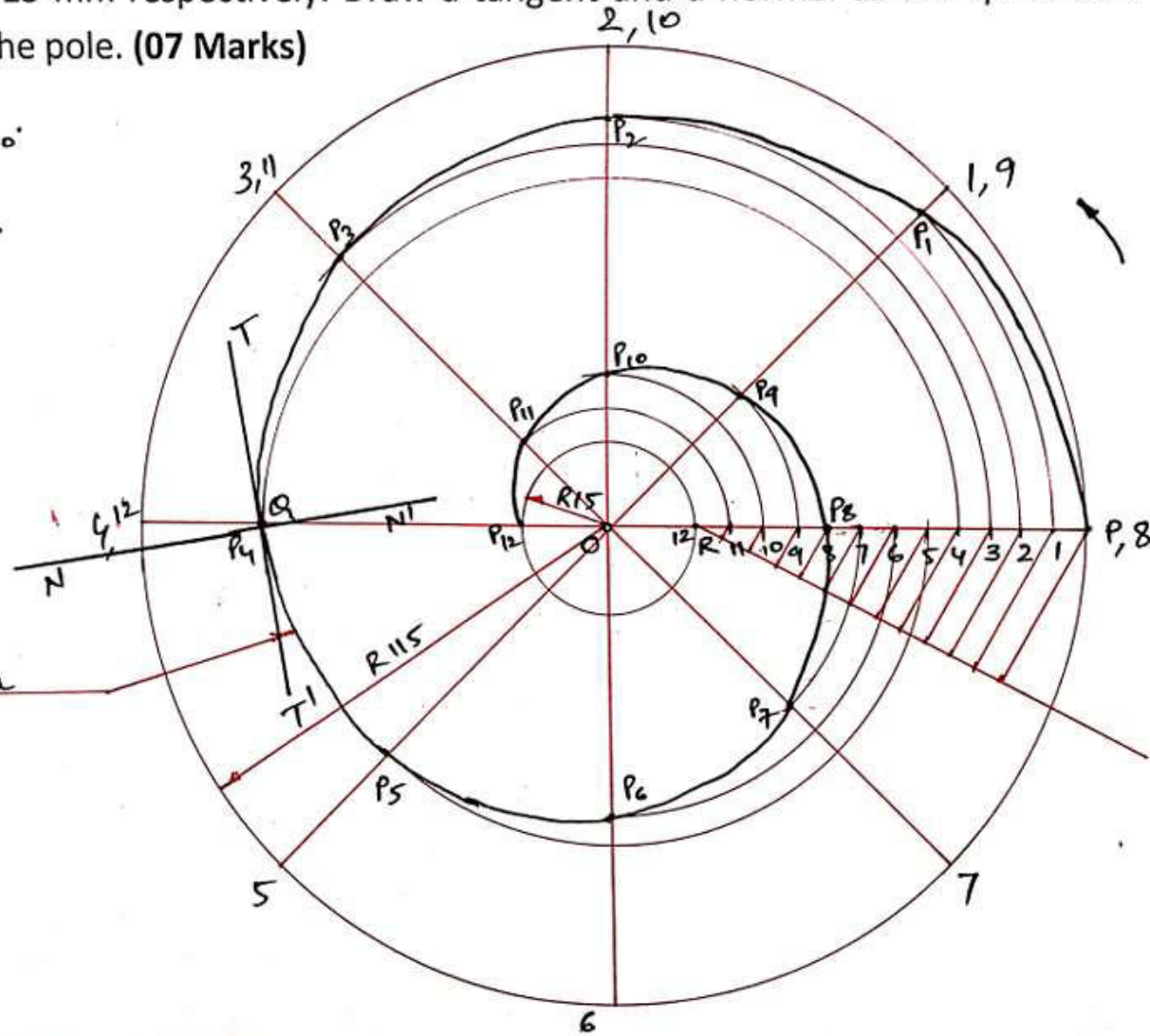
Q. 2 (c) OR Draw an Archimedean spiral of $1\frac{1}{2}$ convolutions, the greatest and least radii being 115 mm and 15 mm respectively. Draw a tangent and a normal to the spiral at a point 63 mm from the pole. (07 Marks)



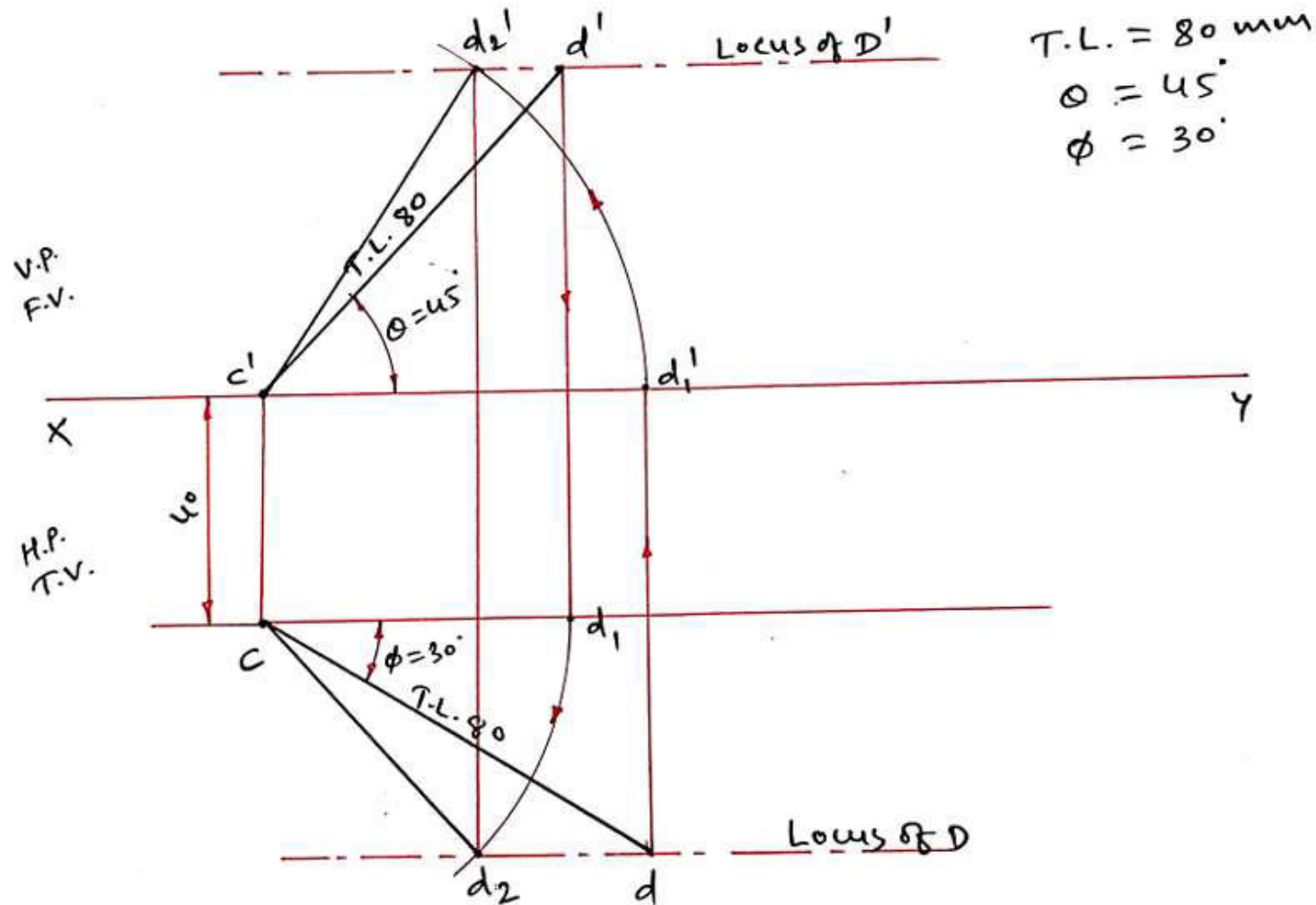
1 Convolution = 360°
 $\therefore 1.5$ convolutions = ?
 $\therefore 1.5 \times 360^\circ = 540^\circ$

$$\therefore \frac{540^\circ}{45^\circ} = 12$$

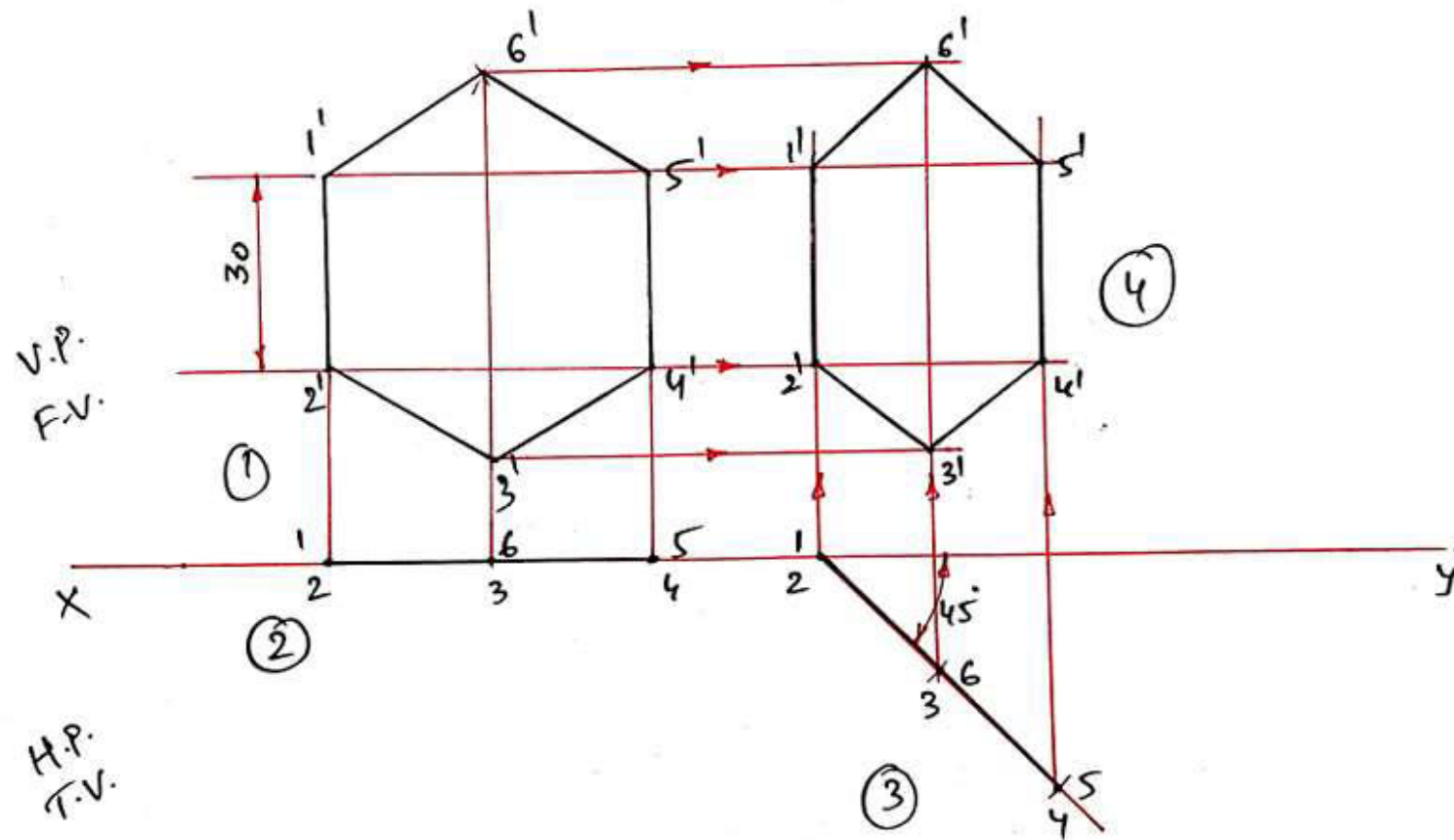
Archimedean
Spiral

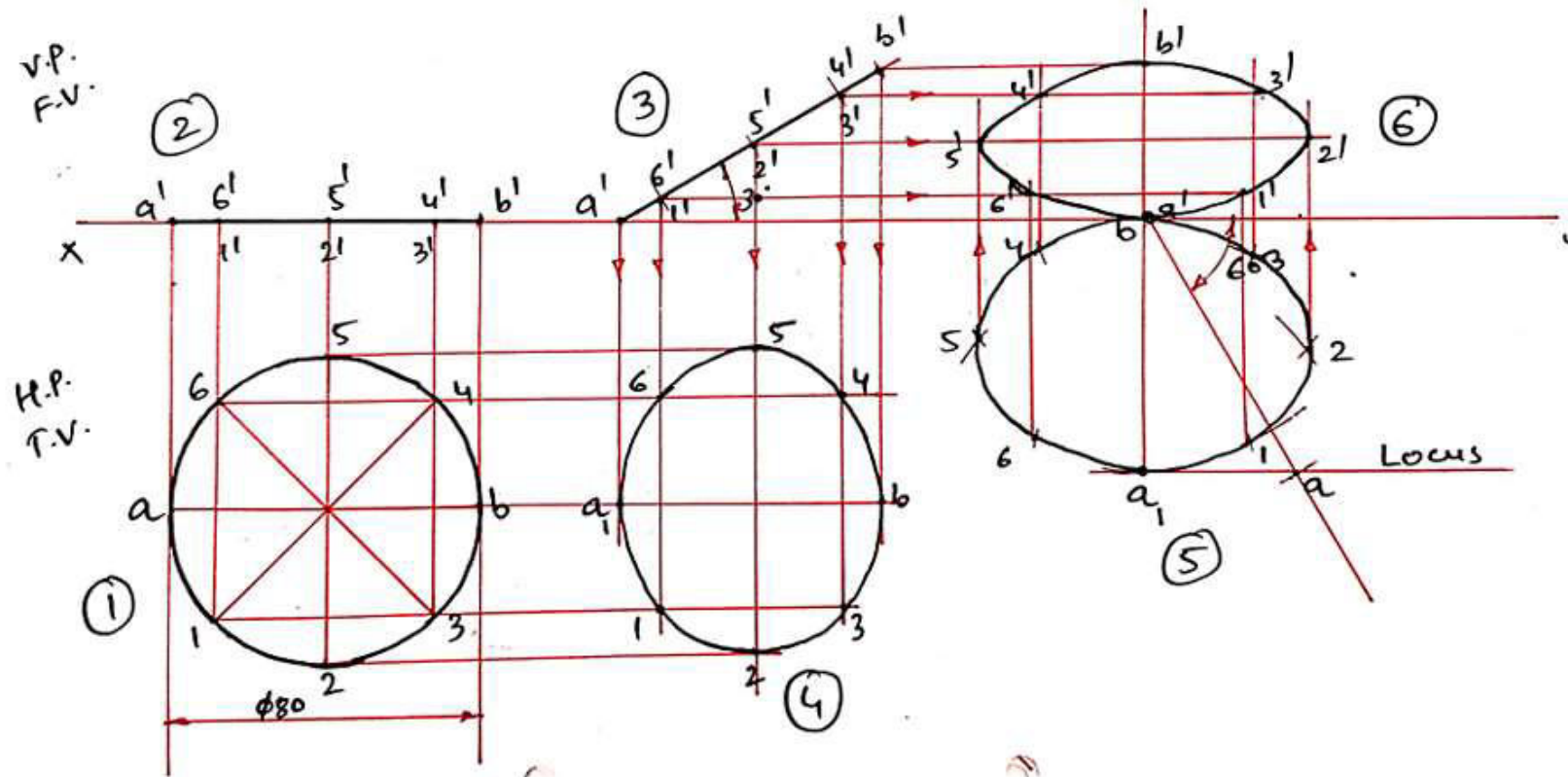


Q.3 (a) A line CD, 80 mm long is inclined at 45° to H.P. and 30° to the V.P., its end C is in H.P. and 40 mm in front of V.P. Draw the projections. **(03 Marks)**



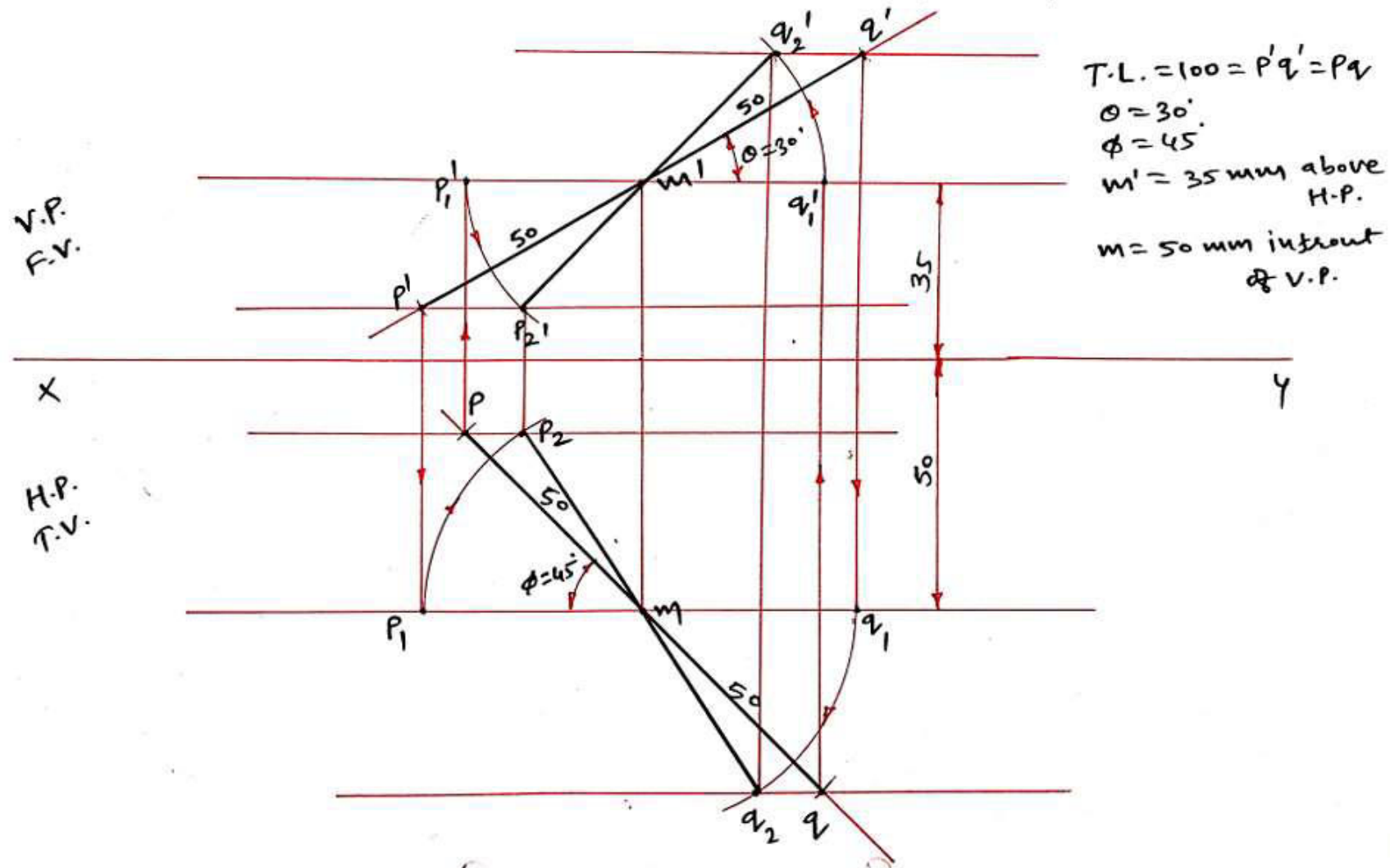
Q. 3 (b) A hexagonal plane of 30 mm side, rests on the V.P. on an edge such that the surface is inclined at 45° to the V.P. Draw its projections. **(04 Marks)**







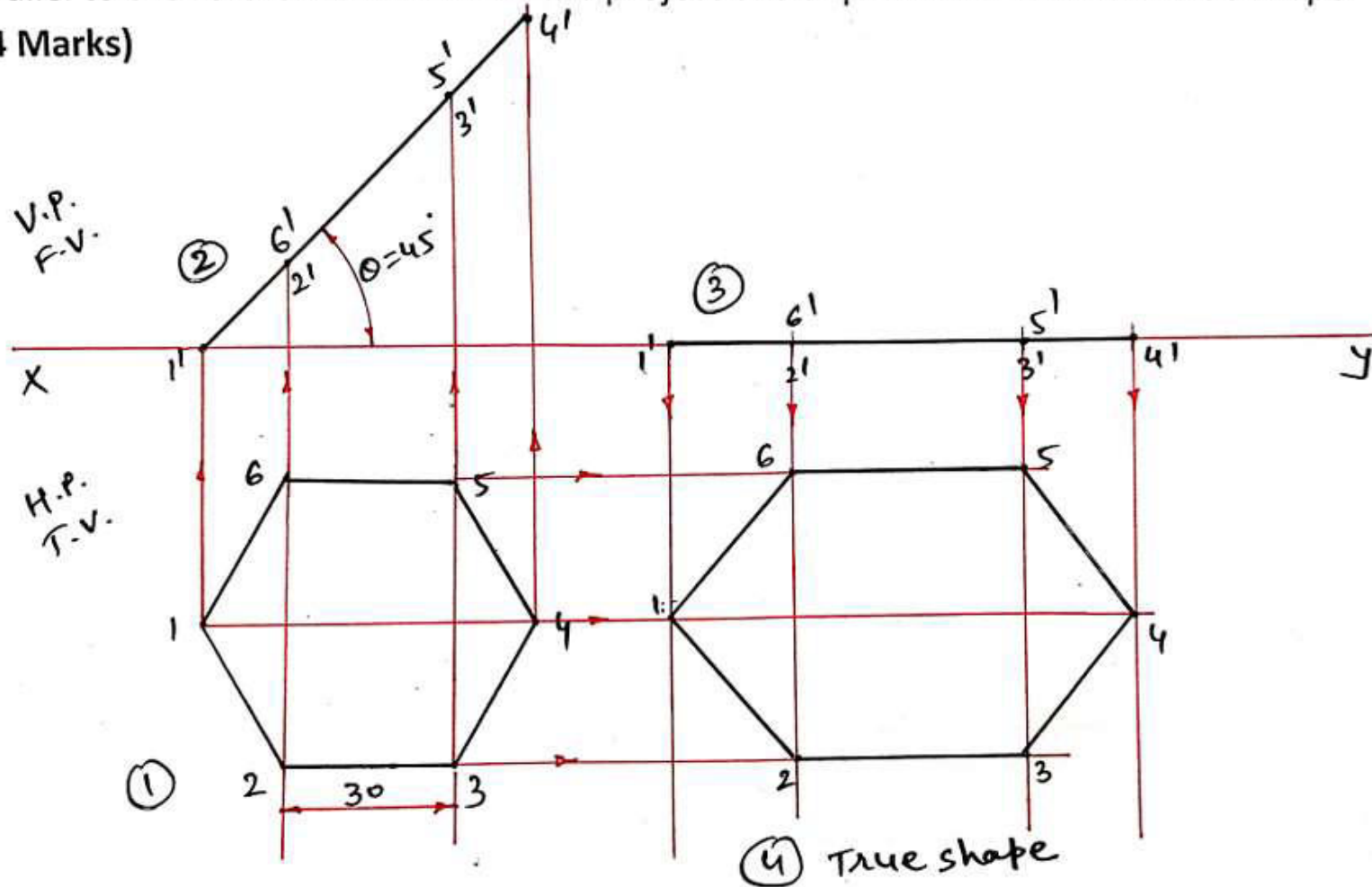
Q. 3 (a) OR A 100 mm long line PQ is inclined at 30° to H.P. and 45° to the V.P. Its mid-point is 35 mm above the H.P. and 50 mm in front of V.P. Draw its projections. (03 Marks)



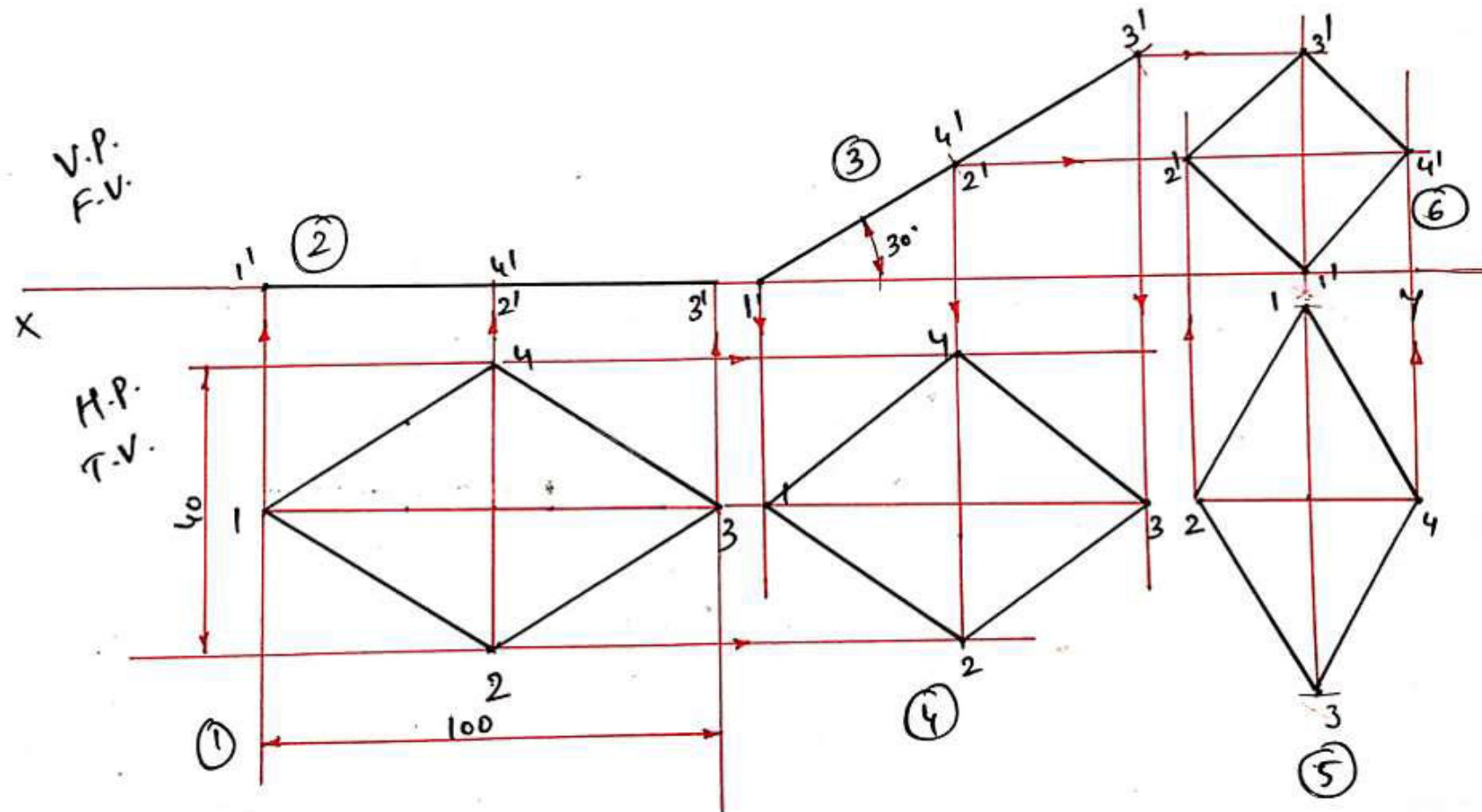


Q. 3 (b) OR The top view of a lamina whose surface is perpendicular to V.P. and inclined at an angle of 45° to H.P. appears as a regular hexagon of 30 mm side, having a side parallel to the reference line. Draw the projections of plane and obtain its true shape.

(04 Marks)

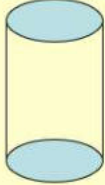
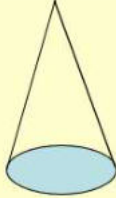
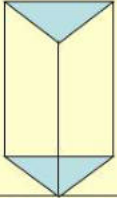
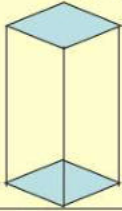
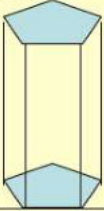
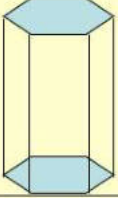
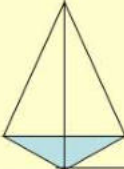
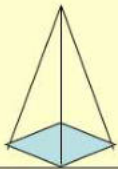
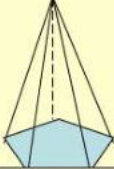
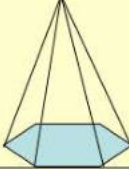
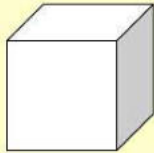
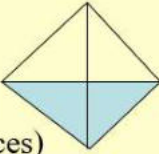


Q.3 (c) OR Draw the projections of a rhombus having 100 mm and 40 mm long diagonals. The bigger diagonal is inclined at 30° to H.P. with one of the end point in H.P. and the smaller diagonal is parallel to both the planes. **(07 Marks)**



Q.4 (a) Classify solids with shapes of drawings. (03 Marks)



Group A				Group B			
Solids having top and base of same shape				Solids having base of some shape and just a point as a top, called apex.			
Cylinder 				Cone 			
Prisms     Triangular Square Pentagonal Hexagonal				Pyramids     Triangular Square Pentagonal Hexagonal			
Cube (A solid having six square faces) 				Tetrahedron (A solid having Four triangular faces) 			

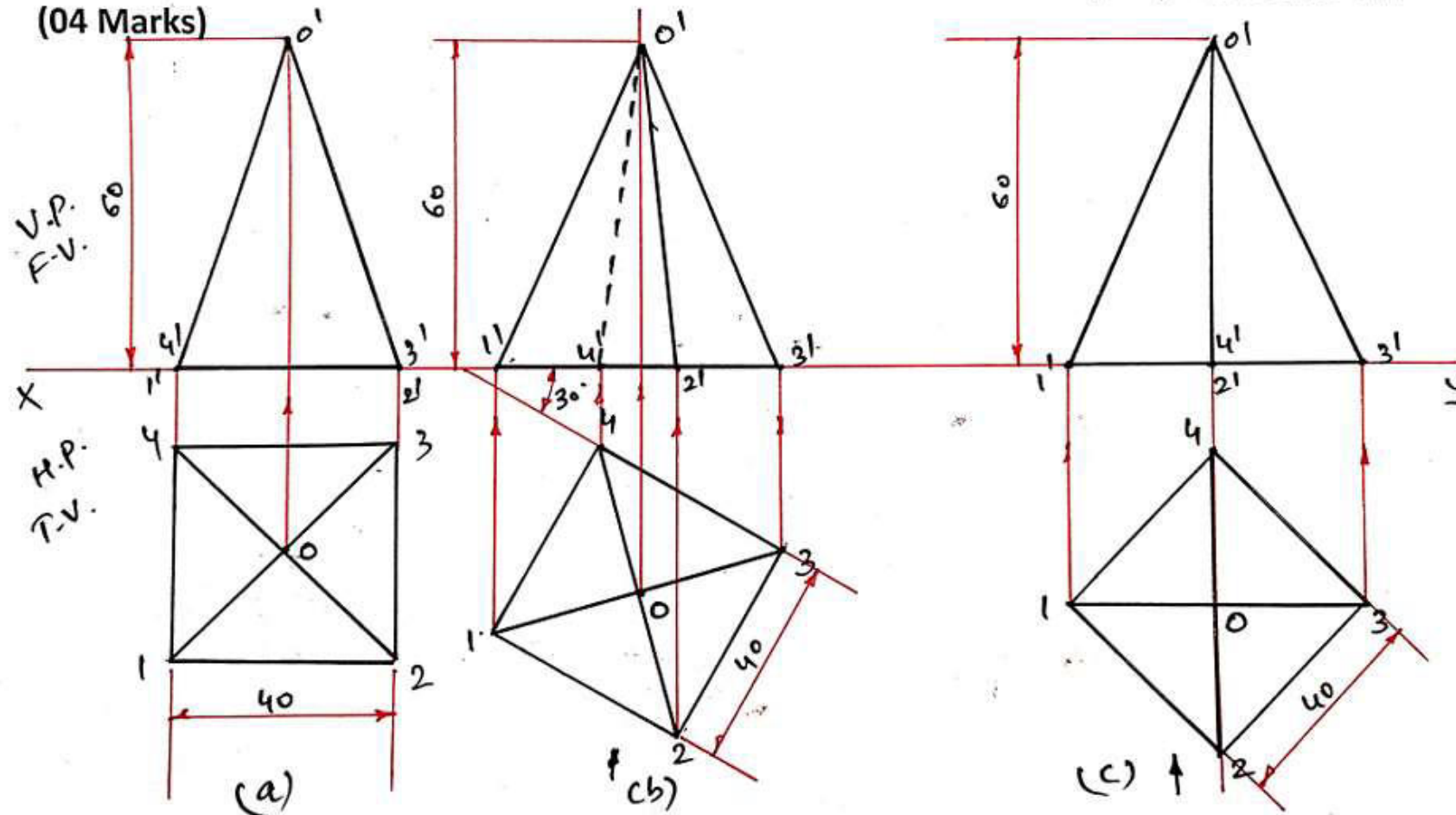
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[Video Link: https://youtu.be/8pFcScIZ1Hs](https://youtu.be/8pFcScIZ1Hs)

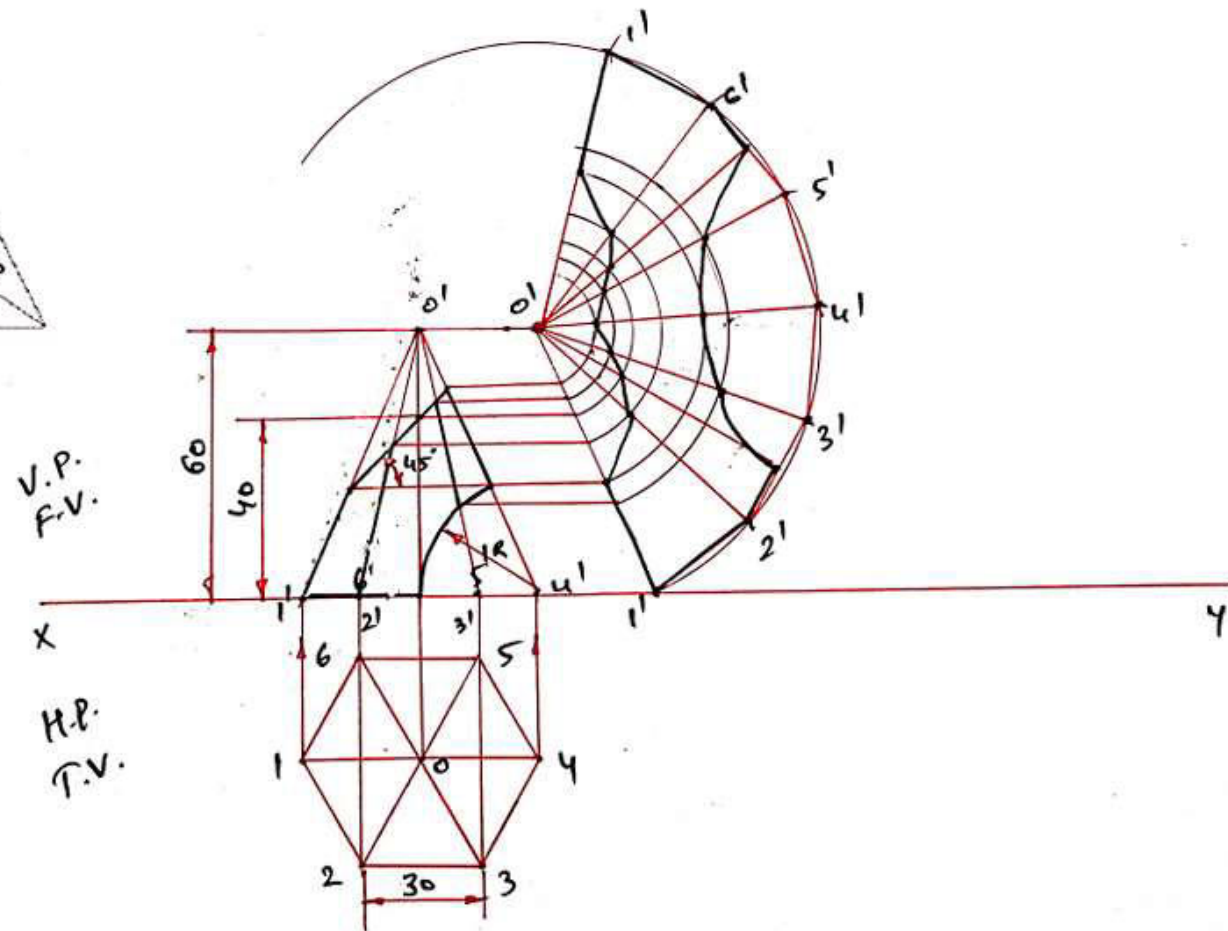
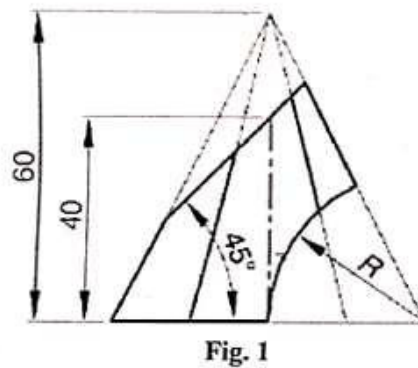


Q. 4 (b) A square pyramid, side of base 40 mm and axis 60 mm is resting on its base on H.P. Draw its projections when (a) a side of the base is parallel to V.P., (b) a side of the base is inclined at 30° to V.P., and (c) all the sides of the base are equally inclined to V.P.

(04 Marks)



Q. 4 (c) A hexagonal pyramid, 30 mm base side and 60 mm long axis rests on the H.P. with a side of base parallel to V.P. It is cut by planes perpendicular to V.P., to obtain the front view as shown in Fig. 1. Draw the development of the lateral surface of the retained solid. **(07 Marks)**





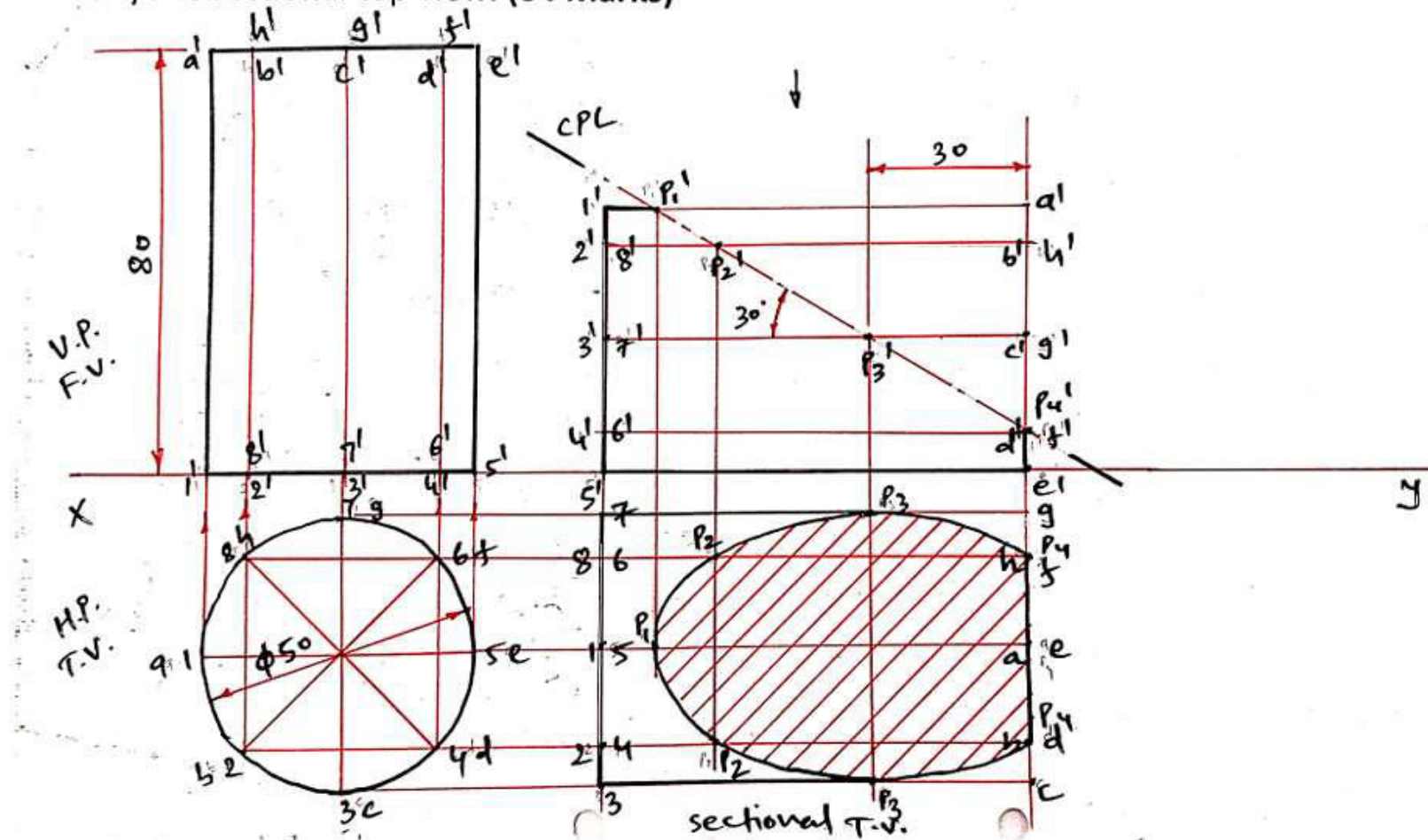
Q. 4 (a) OR Explain difference of any two projection methods. **(03 Marks)**

[Video Link: https://youtu.be/zmAlNodrt1U](https://youtu.be/zmAlNodrt1U)

First Angle Projection Method	Third Angle Projection Method
1. Object is placed in first quadrant	1. Object is placed in third quadrant
2. Object is placed between plane and observer	2. Plane is placed between object and observer
3. Plane should not be considered as transparent	3. Plane should be considered as transparent
4. Front view is drawn above reference line XY	4. Front view is drawn below reference line XY
5. Top view is drawn below the front view	5. Top view is drawn above the front view
6. Left hand side view is drawn right side of front view	6. Left hand side view is drawn left side of front view
7. Right hand side view is drawn left side of front view	7. Right hand side view is drawn right side of front view
8. Bottom view is drawn above the front view.	8. Bottom view is drawn below the front view.
9. Rear view can be drawn beside L.H.S.V. or R.H.S.V.	9. Rear view can be drawn beside L.H.S.V. or R.H.S.V.



Q. 4 (b) OR A cylinder with 50 mm base diameter and 80 mm long axis, is lying on a generator on the H.P. with its axis parallel to the V.P. It is cut by an A.I.P. inclined at 30° to the H.P. and passes through a point on the axis 30 mm from one of its ends. Draw its anyone sectional top view. (04 Marks)



Q. 4 (c) OR Draw any three views using orthographic projections of fig 2. (07 Marks)

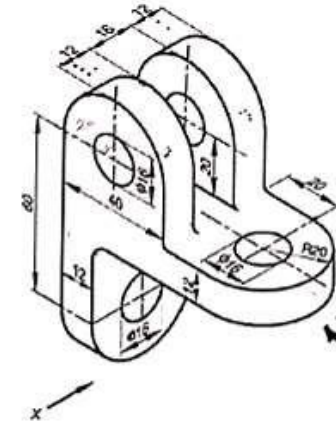


Fig. 2

$$L = 40 + 20 + 20 = 80$$

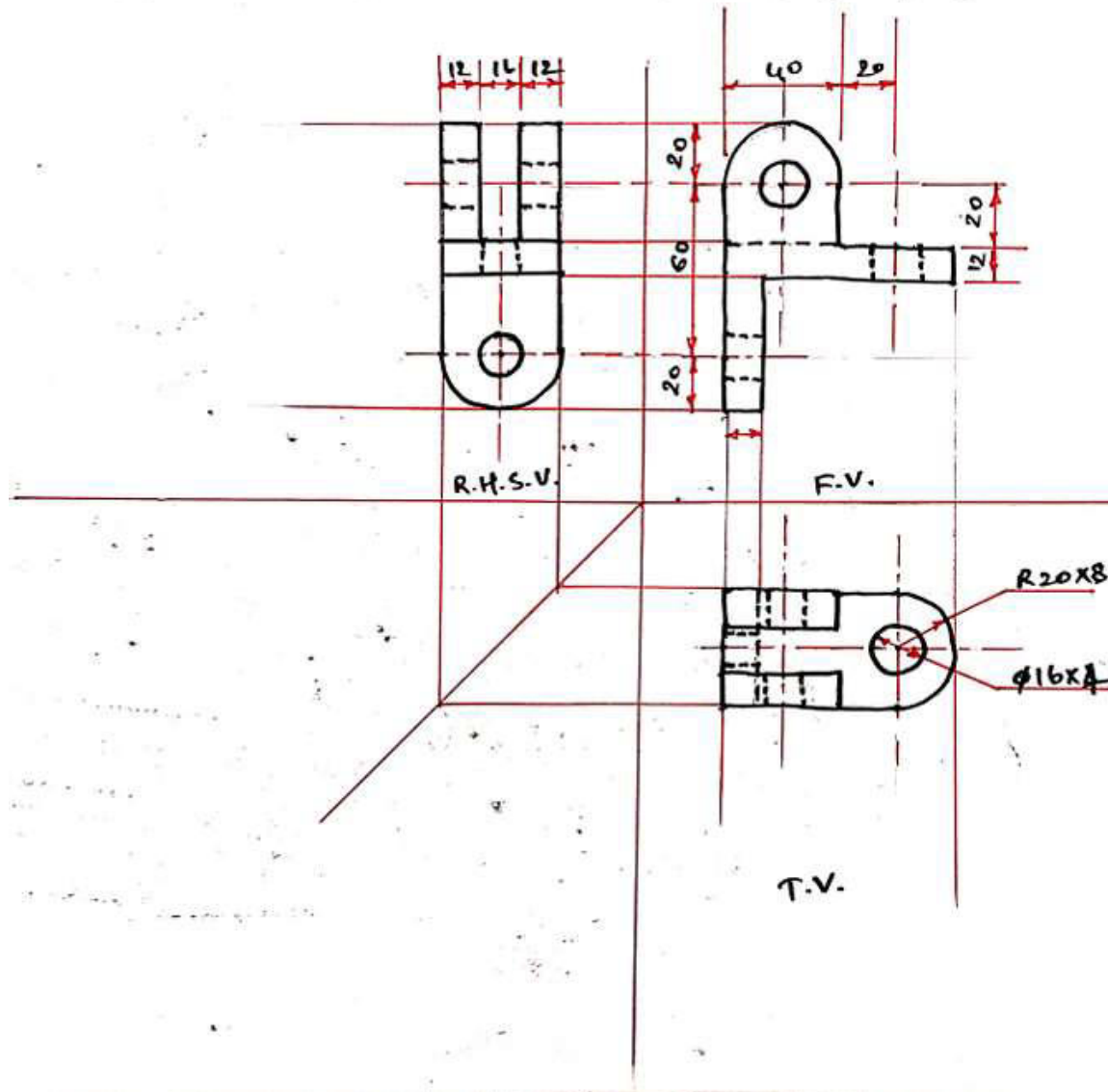
$$H = 60 + 20 + 20 = 100$$

$$W = 12 + 16 + 12 = 40$$

$$F.V. = L + H$$

$$R.H.S.V. = W + H$$

$$T.V. = L + W$$





Q.5 (a) State the importance of AutoCAD in engineering field. (03 Marks)

Importance of AutoCAD is as per the following.

1. With the help of CAD highly accurate and precise drawing can be prepared in less time than manually prepared drawings.
2. Drawing instruments and hand skill is not required.
3. Drawings can be easily saved in computer and it can be modified and copied when necessary.
4. We can print the drawings on printer and plotter when required.
5. No scrap of drawings and no paper wastage.
6. Multicolor display makes visualization more effective.
7. We can increase or decrease the size of drawing to definite scale very easily.
8. Computer aided manufacturing is also possible.



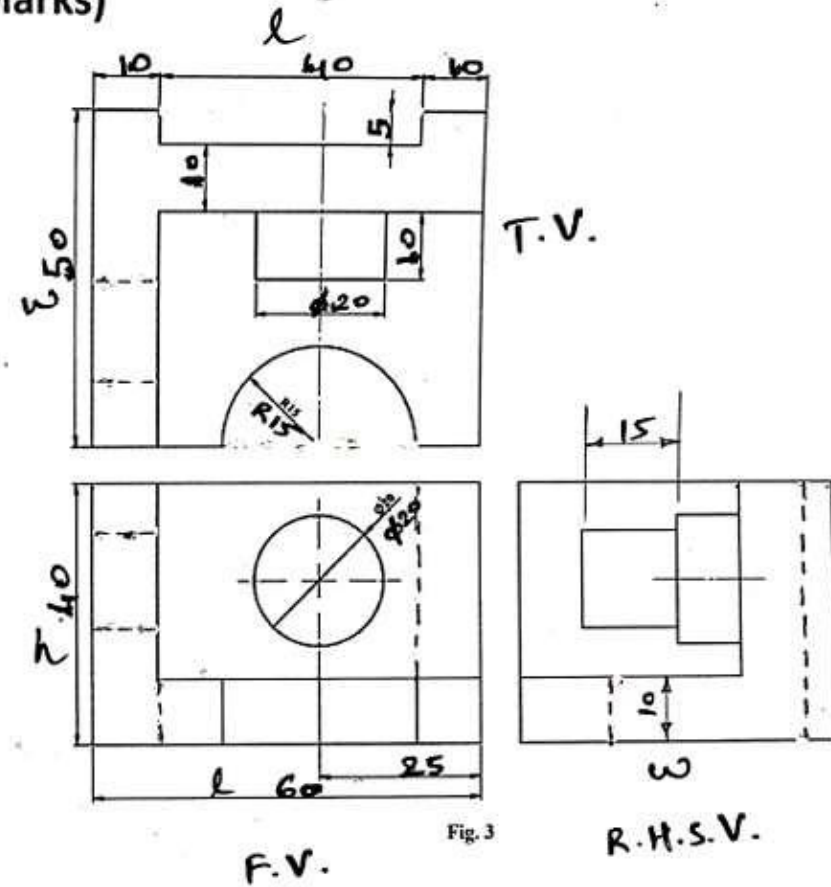
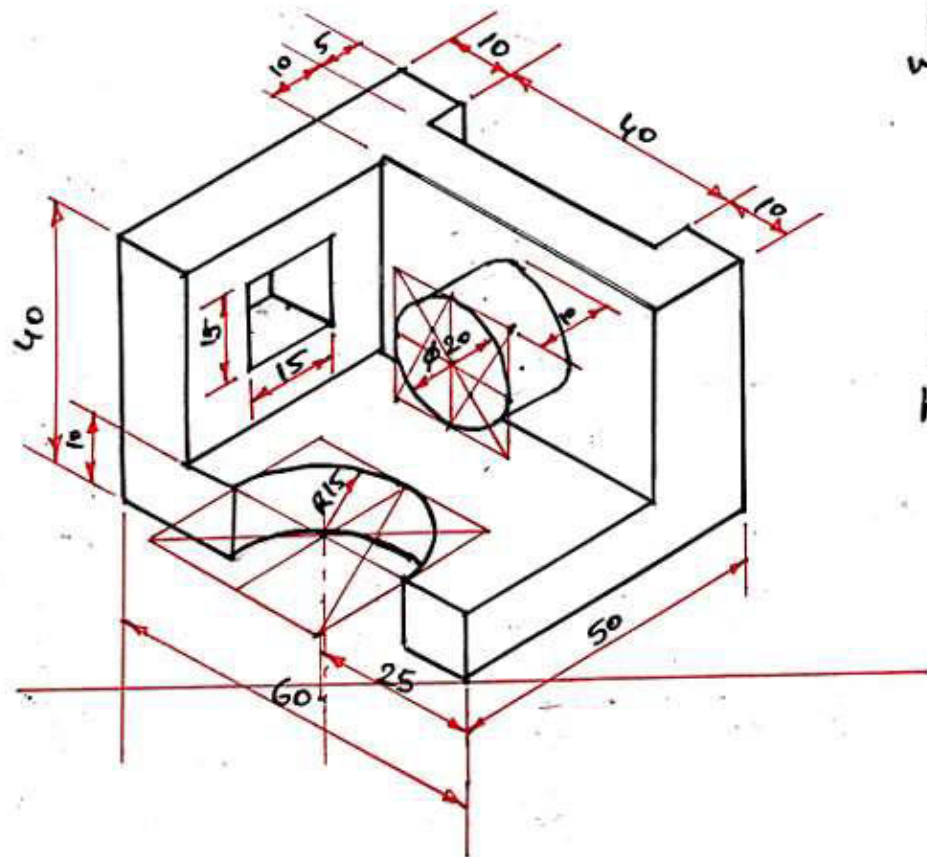
Q.5 (b) Explain the importance of file extension in the communication of drawing.
(03 Marks)

File extensions help the file system and users mainly in two ways.

First, it helps in identifying the type of data it holds.

Second, the file extension allows the operating system to choose the proper application with which to open a file.

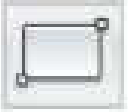
Q. 5 (c) Draw isometric view of given fig. 3. (07 Marks)



length & width = 30'
height = vertical (90')



Q.5 (a) OR Enlist any six CAD drawing preparatory commands with its symbols and function. (03 Marks)

Preparatory Command	Symbol	Function
Line		To draw lines
Rectangle		To draw rectangle of desire size
Polygon		To draw polygon i.e. pentagon, hexagon etc. of desire size
Circle		To draw circle of desire size
Arc		To draw arcs as per requirement
Ellipse		To draw ellipse of desire size



Q.5 (b) OR Identify the command, that used 1) to remove certain unwanted part, 2) to show sectional view, 3) to make a continuous line and 4) to make a circular shape at the edge of drawing. **(04 Marks)**

1. To remove certain unwanted part

Command: Trim

2. To show sectional view

Command: Hatch

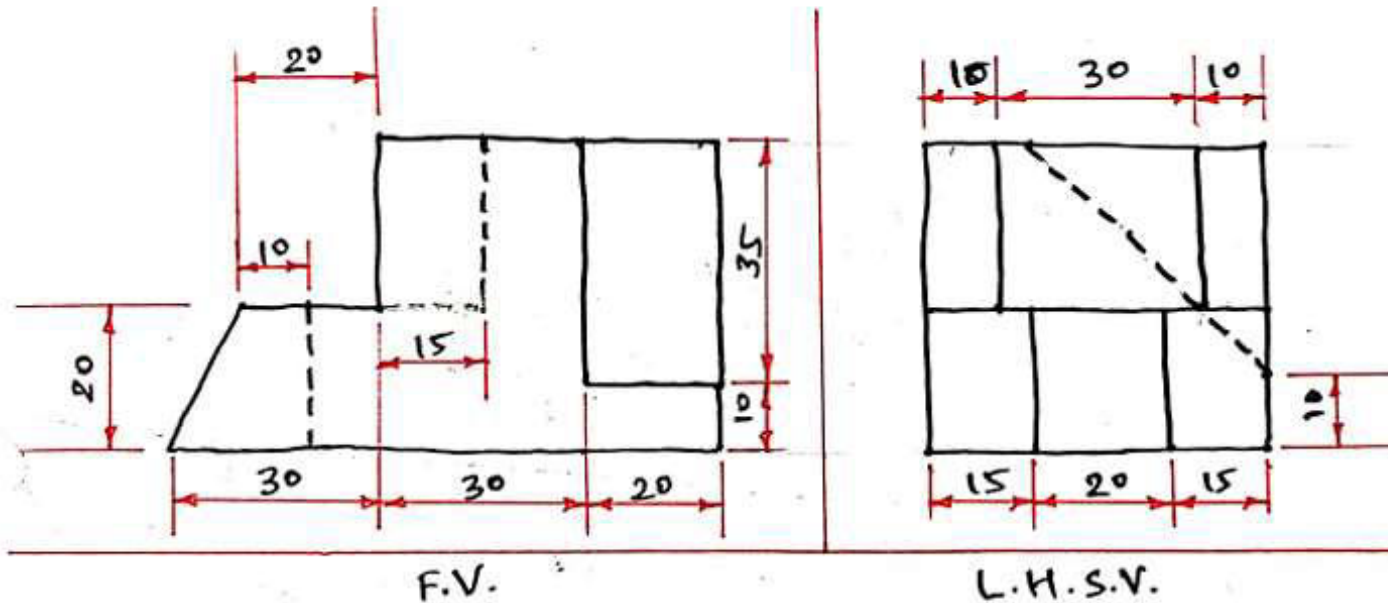
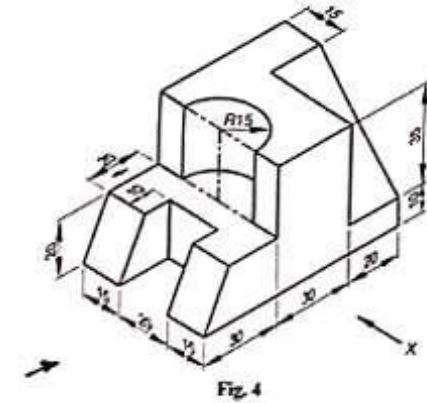
3. To make continuous line

Command: Pline

4. To make circular shape at the edge of drawing

Command: Fillet

Q. 5 (c) OR Draw the Front view of Fig. 4 looking in the X direction and also draw any one side view. (07 Marks)



$$L = 30 + 30 + 20 = 80$$

$$H = 10 + 35 = 45$$

$$W = 15 + 20 + 15 = 50$$

$$F.V. = L + H$$

$$L.H.S.V. = W + H$$